

depend on which shader the materials is using. Shaders are small scripts which contain mathematical calculations and algorithm for calculating the colour of each pixel rendered, which is based on the lighting input and the material configuration.

There are many shaders available within blender which supports the inbuilt cycles render engine & the newly updated EEVEE render engine. Some of the shaders are, Glossy BSDF, Principled BSDF and many more.

The act of building and combining shaders is called shader building, and the act of creating textures for the shaders is called texturing, and the act of UV mapping with texturing and shader building is called surfacing.

MATERIALS AND TEXTURES IN BLENDER

As mentioned above, Materials and textures are what changes your model from being grey and lifeless to colourful and lifelike. You have the ability to add colour, make things glow, make it transparent like a glass or make it look like brick etc.

You must always add a material before you can add texture. To add material, Select the object you want to apply textures too. Then go to the materials panel in the properties window. Then click on the “new” button. You will see more options open up. The material block is used to change some of the physical properties of the object in its appearance. If you're planning on using just straight colour and no texture, this is where you set the object's colour. The panels can float around so they may not be in order. Panels can be collapsed and opened as needed to streamline your view. Some of the basic material panels and settings are:

1) Diffuse

Diffuse is the colour which is given off the object. You will also see settings for the way the material is calculated, and the default is Lambert, and also the intensity slider. The ramp button will allow for a diversity of colour.

By clicking on the colour sample in diffuse, the colour wheel will pop up. You can set the colour using the wheel and the light/dark slider or setting in manually using RGB, HSV or Hex numbers. You can also pick a colour using the eyedropper tool.

2) Specular

Specular setting controls the glossiness of the object. You will see a colour sample, calculation model and ramp similar to the properties of diffuse.